Teske-type hidden Weil descents by field type: where the three

ingredients exist 2026-10-18. A research search round (AGENTS.md "Research searches must leave visual reports"): literature reading plus exact native enumeration over toy fields. Protocol: PROTOCOL.md, frozen at commit `62b4502c7` before the first run. Tool: `examples/teske_variants_field_types.rs` (std-only Rust, 4 threads; `rustc --edition 2021 -O`, tests with `--test`). Recorded runs: `20261018T0630Z` (Parts A and B; results/run-20261018T0630Z/, run report reports/run-20261018T0630Z/run-report.pdf) and `20261018T0845Z` (Part C for every `q`; results/run-20261018T0845Z/, run report reports/run-20261018T0845Z/run-report.pdf), each with `env.txt` beside its files. The first run was killed by a restart of the session's worker while its census was at `q = 13`; the census root search was then replaced by polynomial splitting (byte-identical output for `q ≤ 11`, checked) and Part C was rerun. One earlier launch, `20261018T0624Z`, was stopped after a smoke run exposed a normalisation bug in Part C; its directory keeps a `NOTE.md`. This report as PDF: REPORT.pdf. Conductor is not installed on this host (the installer answers HTTP 403 through the session proxy), so there is no task card. The round solves no discrete logarithm, times nothing, and makes no statement about any deployed curve. Every number below is a count or a formula on exact invariants; the exponents are formula-level labels in the sense of `docs/curves/JACOBIAN_TRANSFERS.md`.

Tags. [T] theorem, stated by a cited source or proved here; [E] exact computation by the tool, reproducible and checked; [H] heuristic; [O] out of scope or not established.

Answer

Teske's trapdoor system (J. Cryptology 19 (2006); ePrint 2003/058) hides a GHS-weak curve over `F_{2^161}` behind an isogeny. It needs three things: a weakness that is **not an isogeny-class invariant**, a **transfer** that follows the isogeny, and a weak set **sparse inside the isogeny class**, so that an auditor cannot find the weak member by walking the class faster than Pollard rho. This round asked over which field types and extension degrees all three exist, and what other ways there are to attach a Weil-descent target to a curve.

- **[T] Prime fields `F_p`: none of the ingredients.** There is no subfield to descend to (GHS 2002 §5), every same-field cover of genus `g ≥ 2` moves the problem into a group of size `p^g` where index calculus costs at least `p^{1−o(1)} > p^{1/2}`, and base change to `F_{p^k}` followed by a descent leaves the subgroup of size `p` inside a group of size `≥ p^k`. The only non-class-invariant structure left, the endomorphism ring, gives no exponential gain. No Teske-type variant via descent exists over `F_p` (§3).
- **[E, T] Binary fields `F_{2^{nl}}`.** The exact combinatorics of `x^n − 1` over `F_2` (Part A) place the sparse regime at `n = 7` for the type-I genus-7 descent only (`δ = 4`, margin `+0.5`, i.e. `+11.5` bits at `N = 161`: an auditor's walk costs as much as rho), while the type-II genus-8 descent (`δ = 3`, `−11.5` bits: the walk costs `2^69`) and Hess's gGHS variant (`δ = 2`) are dense, which is why Menezes–Teske call `GF(q^7)` partially weak. `n = 3` and `n = 5` are dense for every curve (weak fields). The sparse regime reappears at `n = 15, 21, 31` and other odd `n` with small `ord_n(2)`, with genus 15, 21–31, 31, but there the public sub-descents (`n = 5` inside `n = 15`) already beat rho, and `n = 31` is the `F_{2^155}` case of Hess 2003 (`2^{−38}` weak orbit classes per isogeny class). Prime `n` has `m ≥ ord_n(2) + 1` and nothing descends (Menezes–Qu). Table in §4.
- **[E, H] Odd characteristic `F_{q^n}`.** The weak set of a Kummer-descent branch type with `r` points has `≈ q^{r−3}` curves: the exact counts at `q ≤ 13` reproduce the exponent `r − 3` with constants between `0.04` and `2.1` (Part B), so `δ = n + 3 − r` and the margin of the smallest-genus type is `a = n + 3 − r − n/2`: `n = 2: −1` (genus 2, Scholten, no gain anyway), `n = 3: −1/2` (genus 3: dense, the isogeny-walk regime Joux–Vitse conjecture), `n = 4: 0` for the `r = 5` type and `−1` for the `r = 6` types (all genus 5), **`n = 5: +1/2` (genus 5)**, `n = 6: 0` (genus 9), `n = 7: −1/2` (genus 7), prime `n ≥ 11`: genus `≥ 1793` (Diem). **The odd-characteristic analogue of Teske's field is `F_{q^5}`**, with the genus-5 family `y^2 = ∏_{i=0}^{3}(x − α^{q^i})`, `α` of degree 5 (Diem 2003; Hess 2004 Table 3). The isogeny-class statistics (every `j` over `F_{q^n}` traced) show the predicted drift: the share of curves whose class contains a weak member saturates for `n = 3` at the classes with `#E ≡ 0 (mod 4)`, stays flat for `n = 4`, and falls with `q` for `n = 5` (§5).
- **[T, E] A second association, without covering maps.** By Tate, a genus-3 curve `C/F_q` has `Jac(C)` isogenous to `Res_{F_{q^3}/F_q} E` exactly when `L_C(T) = 1 − tT^3 + q^3T^6`. For hyperelliptic `C` this forces `t` even and a Frobenius cycle type `(1,1,3,3)`, `(2,3,3)`, `(1,1,6)` or `(2,6)` on the eight Weierstrass points (Theorem 3, from the Frobenius action on `Jac(C)[2]`), so **no hyperelliptic genus-3 Jacobian is isogenous to the Weil restriction of a prime-order `E/F_{q^3}`, by any isogeny**, which sharpens the remark of Tian 2020 §1 that covering maps from hyperelliptic curves cannot reach prime order. The census of every genus-3 hyperelliptic curve over `F_q`, `q ≤ 13`, finds `9,472` Res-shaped isomorphism classes of which only `331` carry the GHS compositum signature (a Frobenius-cyclic Klein four-group on the hyperelliptic line, Theorem 2): the isogeny relation is far wider than the covering construction. Whether any non-signature class carries a cover of small degree is open [O] (§6).
- **No undiscovered affected curve type was found** inside the enumerated families; the one new structural fact is Theorem 3, and the one new place where a hidden descent can live in odd characteristic is `n = 5`. What remains open, and the next discriminating steps, are listed in §7.

1. Framework: exponents, margin, and what an auditor can do

For `E/F_{q^n}` with a descent target of genus `g` over `F_q` (exponents in units of `log q`, constants dropped):

quantity	exponent	tag and source
Pollard rho on the prime-order subgroup	`n/2`	[T]
index calculus on a genus-`g` Jacobian over `F_q`, double large primes	`2 − 2/g` (`1` for a genus-3 plane quartic)	[H] Gaudry–Thomé–Thériault–Diem; Diem–Thomé 2008; constants grow with `g`
weak curves among the `≈ q^n` isomorphism classes	`q^{n−δ}`	[E] Parts A and B
members of a random isogeny class	`q^{n/2}`	[T] Hasse–Weil; GHS 2002 §5
weak members per class, uniform-distribution heuristic	`q^{n/2−δ}`	[H] GHS 2002 §5; Menezes–Teske 2006; Yee 2016 §5.2 (tested on `F_{2^18}`, `F_{2^24}`, `F_{2^30}`)
auditor's isogeny walk to a weak member, when one exists	`min(δ, n/2)`	[H]
margin `a = δ − n/2`	`> 0`: sparse, hidden; `< 0`: dense, weak field	definition

A hidden descent in Teske's sense needs `a > 0` (the auditor's walk costs at least rho) together with `2 − 2/g < n/2` (the owner gains). With `a < 0` and `2 − 2/g < n/2` the walk plus the descent beat rho for every class that contains a weak member: Menezes–Teske–Weng's weak field, extended by Galbraith–Hess–Smart's isogeny step. The auditor's answer in the sparse regime is not the walk but the direct test of the published curve for every weak type (Part A's magic numbers; Part B's branch types), which costs polynomial time and which the repository's `audit_curve` and `ghs_screen` already do for binary fields.

The exponent `2 − 2/g` is quoted, not measured; MMT 2002 Table 1 and Appendix A give the Enge–Gaudry running times with constants for every composite `N ∈ [100, 600]`, and this round does not redo that.

2. Sources read (exact locations)

source	what it states (quoted or tightly paraphrased)	class
Teske, *An elliptic curve trapdoor system*, ePrint 2003/058 (abstract; the PDF returns HTTP 403 here), J. Cryptology 19 (2006) 115–133	"a pair of elliptic curves over $F_{2^{161}}$": GHS "reduces the ECDLP in $E_s(GF(2^{161}))$ to a hyperelliptic curve DLP in the Jacobian of a curve of genus 7 or 8, which is computationally feasible, but by far not trivial"; "E_{pb} is isogenous to E_s"; "the best attack on the ECDLP in E_{pb} is the parallelized Pollard rho method".	T (abstract only)
Galbraith–Hess–Smart, *Extending the GHS Weil descent attack*, Eurocrypt 2002 (author PDF, Hess's page)	§1: "One might 'hide' such a curve by taking an isogeny to a curve E_1 for which the GHS attack is not effective. Knowledge of the 'trapdoor'…" (the trapdoor idea predates Teske's paper). §5: $p = \min(1, s\, q^{t+1-n/2}/(nr))$ weak classes per isogeny class; $n = 7$: "every curve over F_{q^7} can be attacked by reducing to a hyperelliptic curve of genus 7 or 8 over F_q" for $q \geq 2^{11}$; table of p for $q = 2, \ldots, 2^5$: $0.404, 0.286, 0.269, 0.286, 0.323$; $n = 31$: basic proportion $6q^{-25}$, extended $6q^{-9.5}/(31r)$, "one in 2^{52}" for the IPsec curve; "for curves over large prime fields the techniques of Weil restriction do not apply".	T; the per-class formula is H
Hess, *The GHS attack revisited*, Eurocrypt 2003 (author PDF)	§4: $m = d$ from $b = (\alpha\beta)^2$, genus $2^d - 1$, about $s\, q^{2d-1}/2$ curves; per-class proportion $\min\{1, s\, q^{2d-1}/(2\, q^{n/2}\, n\, r)\}$; $F_{2^{155}}$: 2^{-38} against 2^{-52}; the search costs $\approx 2^{39}$ operations. §5.2 (towers) and §5.3 (descent from extensions: "for prime n this approach will however not give an improvement").	T; proportions H
Hess, *Weil descent attacks* (survey, author PDF), Thm 30 (from Diem 2003) and Table 3	Odd characteristic, prime n: $g \geq 2^{\varphi_2(n)-2}(n-4)+1$; "For $n = 5$ and $n = 7$ there are families … for which g assumes the lower bounds 5 and 7"; $n = 11, 13, 17, 19, 23$: $g \geq 1793, 9217, 833, 983041, 9729$; "The attack is not feasible for prime $n \geq 11$". Table 3 lists the $n = 5$ family $Y^2 = \prod_{i\in\{0,1,2,3\}}(X - \sigma^i(a))$ and the $n = 7$ family $\prod_{i\in\{0,1,2,4\}}$.	T
Maurer–Menezes–Teske, LMS JCM 5 (2002) (Cambridge PDF)	Table 1: $N = 161, n = 7, l = 23, m = 4, g = 7, I = 94, \rho = 80, T = 34$; $N = 165, n = 15, m = 5, g = 15, I = 57$; $N = 170, n = 5, m = 5, g = 15, I = 171$. Remark 20: "only $\approx 2^{94}$ out of $\approx 2^{162}$ elliptic curves over $F_{2^{161}}$ have magic number $m = 4$ relative to $n = 7$"; finding an isogenous $m = 5$ curve over $F_{2^{165}}$ "may be difficult, as one has essentially to search through the entire set of 2^{57} curves". §7: the random curve E161-2 has $(m, g) = (7, 64)$ for $(n, l) = (7, 23)$.	T
Menezes–Teske, *Cryptographic implications of Hess' generalized GHS attack*, ePrint 2004/235 (abstract)	"$GF(q^7)$ are potentially partially weak … for half of all elliptic curves over $GF(q^7)$, namely those curves E for which $\#E$ is divisible by 4"; $GF(q^3)$ partially weak, $GF(q^6)$ potentially weak, $GF(q^8)$ potentially partially weak; N not divisible by 3, 5, 6, 7, 8: not weak.	T (abstract)
Menezes–Teske–Weng, *Weak fields for ECC*, CT-RSA 2004 (ePrint 2003/128, abstract)	fields such as $GF(2^{210})$ are weak: "all instances … can be solved in significantly less time" than rho.	T (abstract)
Karabina–Menezes–Pomerance–Shparlinski, CACR 2010-03	eq. (4): gGHS genus $g = 2^t - 2^{t-r} - 2^{t-s} + 1$; Thm 9: for all but $o(q^n)$ of the b, $g \geq 2^{n/2+o(n)}$; "the gGHS attack can be useful for at most $q^{16 \log_2 n + O(1)}$ elliptic curves"; with isogenies "at most $q^{n/2+O(\log n)}$ … negligible".	T
Yee, *On the effectiveness of isogeny walks for extending cover attacks*, MMath thesis, Waterloo 2016 (PDF)	§5.2: weak orbit classes per isogeny class $s\, q^{t+1-n/2}/(nr)$; $n = 3$: $2(q^2-1)$ weak curves, $\sqrt{q/(nr)}$ per class, "around half of the ordinary isogeny classes will not contain any weak curves"; §6.2: $F_{2^{150}}$ "partially weak" (walk 2^{50} steps, 2^9 faster than rho per step); §7: for odd characteristic "an effective method for discerning whether a curve is weak will be needed in order to utilize the isogeny walk in a meaningful way".	T, H as labelled
Diem, Compositio 147 (2011) (Cambridge PDF)	Prop. 4.10: $g(C) \leq (2^n - 1)(2n - 1)$ under Condition 2.7; "Let q be odd. In this case $k(E)\|k(x)$ is (tamely) ramified at 4 places."	T
Chao (with Momose), ECC 2007 slides (pdf)	sl. 20–21: $(2,\ldots,2)$ coverings with $g = d\, g_0$ exist in odd characteristic only for $(d, n) = (2,2), (5,4), d = 2^n - 1$, and decomposable $(3,3)$; densities $\theta(q^2)$ for $d = 3$, $\theta(q^2)$ for $(5,4)$, $\theta(q^{d\ell-3})$? for $d = 2^n - 1$; sl. 24 (hyperelliptic genus-3 family, degree 8), sl. 25 (non-hyperelliptic Types 1–2, degree 4), sl. 33: Type I $\#\{\lambda\} \approx q^{3/2}$.	T; densities marked "?" are H
Iijima–Momose–Chao, ePrint 2013/487 and 2015/805 (abstracts)	complete classification of $(2,\ldots,2)$-coverings under the isogeny condition $g(C) = d \cdot g(C_0)$ for elliptic and genus-2, 3 hyper-	T (abstracts)

	elliptic C_0; "17 classes are newly found"; for elliptic C_0 over k_d with d = 2, 3, 5, 7.	
Joux–Vitse, Eurocrypt 2012 (author PDF)	§3: two isogeny-walk strategies; §4.1: the family (3), θ(q^2) curves, walk ≈ q steps, "we conjecture that all curves with cardinality divisible by 4 are vulnerable"; (4): non-hyperelliptic covers for "about half of the curves having their full 2-torsion defined over F_{q^3}"; §5: F_{p^4}, "approximately 3/4 of the elliptic curves … are directly vulnerable" to a genus-2 cover over F_{p^2}, most admit a genus-9 cover over F_p; App. C: n = 6, genus 9 for "at most 1/q^3" of the curves.	T; conjecture H
Tian, *Cover attacks for elliptic curves with prime order*, arXiv 2012.07173 (2020; J. Cryptology 36 (2023))	abstract: an F_q-rational (ℓ,ℓ,ℓ)-isogeny from Res(E) to the Jacobian of a genus-3 curve, then index calculus, Õ(q) "in some elliptic curve groups of prime order over F_{q^3}"; §1: "a necessary condition is that the elliptic curve has three rational points of order 2 [3,13]"; "[19] shows that C needs to be non-hyperelliptic if the elliptic curve has prime order"; the isogeny needs an F_q-rational maximal isotropic Λ ⊂ W[ℓ] not containing subgroups of E, E^σ, E^{σ^2}; example over F_{523^3} with prime order.	T
Gorla–Massierer, arXiv 1405.1059	trace-zero index calculus Õ(q^{2−2/(n−1)}) for E/F_q seen over F_{q^n}, "lower than that of generic attacks … for n ≥ 5"; feasible for n = 3 up to 30-bit q; for n = 5 the Gröbner step is the bottleneck. A class-invariant attack (it depends only on the group), so not a trapdoor.	T
Arita, Asiacrypt 2000 (abstract, Springer page)	some curves over F_{3^n} of composite degree are weaker than rho via a C_ab curve on the Weil restriction.	T (abstract)
Arita–Matsuo–Nagao–Shimura, IEICE E89-A (2006) (abstract)	quartic extensions of odd characteristic: reduce to a Scholten form, then genus 2 over F_{q^2}, then a C_ab model over F_q; "many" such curves.	T (abstract)

Not obtained (HTTP 403 or paywall): the ePrint PDFs of Teske 2003/058, Menezes–Teske 2004/235, MTW 2003/128, Momose–Chao 2009/236, Iijima–Momose–Chao 2009/613, 2013/487, 2015/805, the IACR archive PDFs of GHS 2002 and Hess 2003 (the author copies were used instead), Diem's JRMS 2003 paper, Thériault's JRMS 2003 paper, Scholten's preprint (404). Nothing below rests on their bodies except as reported by the papers that cite them.

3. Prime fields: no descent variant [T]

Let E/F_p with a prime-order subgroup of size r ≈ p. Rho costs p^{1/2}.

- **No Weil restriction.** F_p has no proper subfield, so there is no extension K/k to descend through: the GHS, gGHS, Kummer and Artin–Schreier constructions all start from [K : k] = n ≥ 2 (GHS 2002 §5; Hess survey §5).
- **Same-field covers do not pay.** A cover C → E over F_p with C of genus g ≥ 2 (the repository's COVERS.md constructs genus-2 ones for every E/F_p) moves the problem into Jac(C)(F_p) of size ≈ p^g. Index calculus there costs Õ(p^{2−2/g}) ≥ Õ(p) (double large primes; Diem's plane-curve Õ(p^{2−2/(d−2)}) is ≥ Õ(p) as well), above p^{1/2}. Genus 1 covers are isogenies and change nothing.
- **Base change then descend does not pay.** Over F_{p^k} the subgroup is still of size p inside E(F_{p^k}) of size p^k; any descent to a Jacobian over F_p of genus ≥ k lands in a group of size ≥ p^k, with the same bound. JACOBIAN_TRANSFERS.md prices this route at the most favourable genus and finds no formula_advantage row for a prime-field curve.
- **What varies inside a prime-field isogeny class.** The trace, the group order, the embedding degree and anomaly are class invariants and cannot be hidden. The endomorphism ring and the rational torsion structure vary with the position in the volcano, but the only ECDLP effect known is the automorphism factor √#Aut, visible on the curve itself (j ∈ {0, 1728}). research/nist_trapdoor_slot_program_20261011/ (hypothesis H1d) already records that the isogeny-hiding *asymmetry* exists over F_p while no weak class exists to hide.

So the Teske pattern has no prime-field variant through descent; the repository's P-256 assessment stands. [O] A weakness over F_p that is neither a class invariant nor a cover would be new mathematics; none is proposed here.

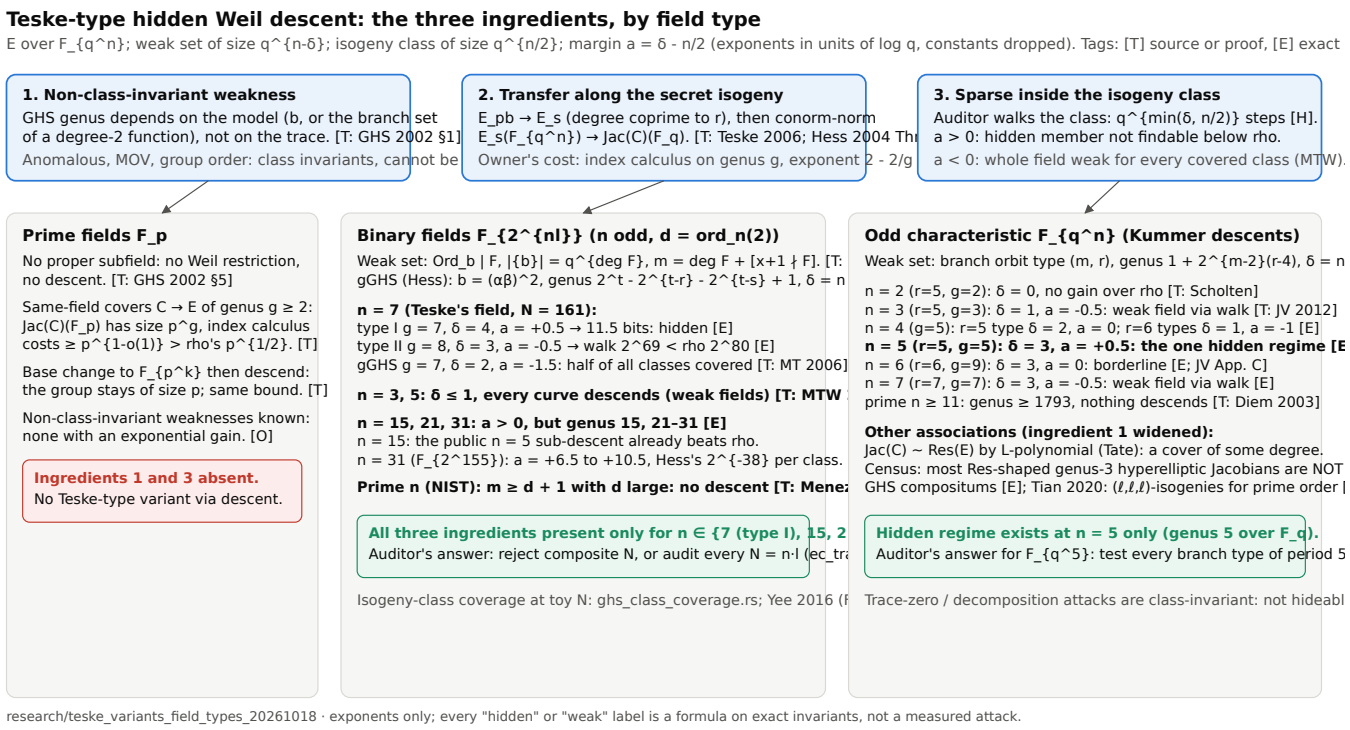


Figure 1: Taxonomy of the three ingredients by field type (figures/taxonomy.svg).

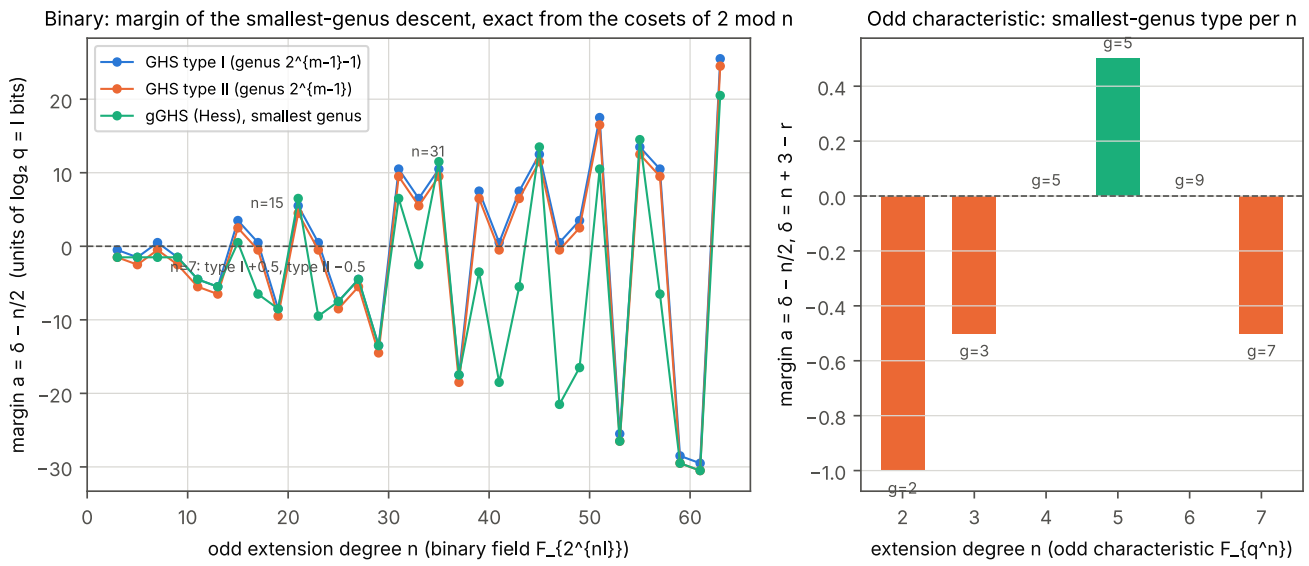


Figure 2: Margins by n (figures/margins.svg, from binary_summary.csv and oddchar_types.csv).

4. Binary fields: the exact map of magic numbers (Part A) [E, T]

For odd `n` the irreducible factors of `x^n − 1` over `F_2` are the cyclotomic cosets of 2 modulo `n`; a product `F` of them with `lcm(orders) = n` is the annihilator of a `b` that lies in no proper subfield, the number of such `b` is exactly `q^{deg F}` (`q = 2^l`), and the magic number and genus are `m = deg F + [x+1 |/ F]`, `g = 2^{m−1} − [x+1 |/ F]` (GHS Thm 1; MMT Thm 5). Hess's gGHS pairs `(F_α, F_β)` give genus `2^t − 2^{t−r} − 2^{t−s} + 1` from ≈ `q^{r+s−1}` values of `b`. The tool enumerates all of this for `3 ≤ n ≤ 99` (`binary_types.csv`, `binary_gghs.csv`, `binary_summary.csv`) and reproduces, as unit tests, the GHS 2002 per-class table for `n = 7`, Hess's `2^{−52}` and `2^{−38}` at `n = 31`, and MMT's counts `2^94`, `2^57`, `2^171` for `N = 161, 165, 170` (protocol checks 1–4).

Margins of the smallest-genus descent (`δ = n − deg F`; margin `a = δ − n/2`; `binary_summary.csv`):

n	ord_n(2)	type I: (deg, g, δ)	a_I	type II: (deg, g, δ)	a_II	gGHS (r+s, g, δ)	a_gGHS
3	2	(2, 3, 1)	−0.5	(3, 4, 0)	−1.5	(2+2, 3, 0)	−1.5
5	4	(4, 15, 1)	−1.5	(5, 16, 0)	−2.5	(1+4, 15, 1)	−1.5
7	3	**(3, 7, 4)**	**+0.5**	(4, 8, 3)	−0.5	(3+3, 7, 2)	−1.5
9	6	(6, 63, 3)	−1.5	(7, 64, 2)	−2.5	(1+6, 63, 3)	−1.5
11	10	(10, 1023, 1)	−4.5	(11, 1024, 0)	−5.5	—	−4.5

13	12	(12, 4095, 1)	−5.5	(13, 4096, 0)	−6.5	—	−5.5
15	4	**(4, 15, 11)**	**+3.5**	(5, 16, 10)	+2.5	(4+4, 15, 8)	+0.5
17	8	(8, 255, 9)	+0.5	(9, 256, 8)	−0.5	(8+8, 255, 2)	−6.5
19	18	(18, 2^18−1, 1)	−8.5	(19, 2^18, 0)	−9.5	—	−8.5
21	6	**(5, 31, 16)**	**+5.5**	(6, 32, 15)	+4.5	**(2+3, 21, 17)**	**+6.5**
23	11	(11, 2047, 12)	+0.5	(12, 2048, 11)	−0.5	(11+11, 2047, 2)	−9.5
25	20	(20, 2^20−1, 5)	−7.5	(21, 2^20, 4)	−8.5	—	−7.5
27	18	(18, 2^18−1, 9)	−4.5	(19, 2^18, 8)	−5.5	—	−4.5
31	5	**(5, 31, 26)**	**+10.5**	(6, 32, 25)	+9.5	(5+5, 31, 22)	+6.5
33	10	(10, 1023, 23)	+6.5	(11, 1024, 22)	+5.5	(10+10, 1023, 14)	−2.5
35	12	(7, 127, 28)	+10.5	(8, 128, 27)	+9.5	**(3+4, 105, 29)**	**+11.5**
45	12	(10, 1023, 35)	+12.5	(11, 1024, 34)	+11.5	(4+6, 945, 36)	+13.5
63	6	(6, 63, 57)	+25.5	(7, 64, 56)	+24.5	(6+6, 63, 52)	+20.5
93	10	(7, 127, 86)	+39.5	(8, 128, 85)	+38.5	(2+5, 93, 87)	+40.5

Reading. (i) For prime `n`, `deg F ≥ ord_n(2)` and the sparse regime (`a > 0`) coincides with a genus that is useless (`n = 17, 23`: genus 255, 2047), except `n = 7` (genus 7) and `n = 31` (genus 31): Teske's field and Hess's `F_{2^155}`. (ii) For composite `n` a product of *non-primitive* factors can beat the primitive one: `n = 21` reaches `m = 6` (genus 31) with `(x^2+x+1)(x^3+x+1)` instead of `m = 7` from an order-21 sextic, and Hess's pairing lowers it to genus 21; `n = 35, 45, 93, 99` behave alike (check 4). (iii) `n = 3, 5` have `δ ≤ 1` for every variant: the descent is public and the field is weak (MTW 2004; Menezes–Teske 2006 "`GF(q^3)` partially weak"). (iv) At `n = 7` the three variants split: type I (genus 7, `δ = 4`) is sparse, type II (genus 8, `δ = 3`) and gGHS (genus 7, `δ = 2`) are dense. The GHS 2002 table (`p = 0.404, …, 0.323` for `q = 2, …, 2^5`) and the "`q ≥ 2^11`: every class" statement are the type-II numbers; Menezes–Teske's "half of all curves" (`#E ≡ 0 mod 4`) is the gGHS number.

Bits for Teske's and the literature's fields (`binary_bits.csv`; rho `N/2`; owner `(2 − 2/g)·l`; walk `min(δ, n/2)·l`; margin `(δ − n/2)·l`; "per class" = expected weak Frobenius-orbit classes per isogeny class, capped at 1):

`N = n·l`	variant	`g`	`δ`	rho	owner	walk	margin	`log2 #weak`	per class
161 = 7·23	GHS-I	7	4	80.5	39.4	80.5	+11.5	71	4.3e−6
161 = 7·23	GHS-II	8	3	80.5	40.2	69.0	−11.5	94	1
161 = 7·23	gGHS 3+3	7	2	80.5	39.4	46.0	−34.5	117	1
161 = 7·23	generic	64	0	80.5	45.3	0	−80.5	162	1
161 = 23·7	GHS-I	2047	12	80.5	14.0	80.5	+3.5	79	1.1e−3
155 = 5·31	GHS-I	15	1	77.5	57.9	31.0	−46.5	125	1
155 = 5·31	generic	16	0	77.5	58.1	0	−77.5	156	1
155 = 31·5	GHS-II	32	25	77.5	9.7	77.5	+47.5	33.6	1.9e−16
155 = 31·5	gGHS 5+5	31	22	77.5	9.7	77.5	+32.5	48.6	6.4e−12
165 = 5·33	GHS-I	15	1	82.5	61.6	33.0	−49.5	133	1
165 = 15·11	GHS-I	15	11	82.5	20.5	82.5	+38.5	46	3.1e−14
165 = 15·11	GHS-II	16	10	82.5	20.6	82.5	+27.5	57	6.4e−11
168 = 7·24	GHS-I	7	4	84.0	41.1	84.0	+12.0	74	2.9e−6
168 = 21·8	gGHS 2+3	21	17	84.0	15.2	84.0	+52.0	35	5.3e−18
210 = 5·42	GHS-I	15	1	105	78.4	42.0	−63.0	169	1
210 = 7·30	GHS-I	7	4	105	51.4	105	+15.0	92	2.9e−7
210 = 7·30	GHS-II	8	3	105	52.5	90.0	−15.0	122	1
210 = 15·14	GHS-I	15	11	105	26.1	105	+49.0	58	1.7e−17
210 = 21·10	gGHS 2+3	21	17	105	19.0	105	+65.0	43	5.2e−22
300 = 15·20	GHS-I	15	11	150	37.3	150	+70.0	82	5.6e−24
600 = 15·40	GHS-I	15	11	300	74.7	300	+140	162	2.4e−45

What the table says, and what it does not. [E] At `N = 161` the two genera of Teske's abstract are two different auditor problems: a planted genus-7 (type I) curve costs the auditor a walk as long as rho (the class holds `4·10^{−6}` such orbit classes on average, so one must be planted), while a genus-8 (type II) curve is reached by a `2^69`-step walk, the figure MMT 2002 Remark 20 and Teske's paper discuss; the gGHS genus-7 route covers

half of all classes outright (Menezes–Teske 2006). [H] The owner's 39–40 bits are the formula $(2 - 2/g)\cdot l$; MMT's Enge–Gaudry estimate for the same genus is $T = 34$ bits (Table 1), the same order. [E] At $N = 155$ the public $n = 5$ descent (genus 15/16 over $F_{2^{31}}$, 58 bits) already sits below rho for every curve, and the sparse $n = 31$ descents (genus 31–32 over F_{2^5}) are the ones GHS 2002 searched 31 days for and did not find for the IPsec curve; Hess's 2^{-38} per class is the gGHS row. [E] For $N = 165, 210, 300, 600$ the $n = 15$ type-I descent (genus 15 over $F_{2^{N/15}}$) is sparse by 27–140 bits while the public $n = 5$ descent is already below rho: an auditor who only prices the public route misses the sparse one by 40–150 bits of owner advantage. [O] Nothing here is a running time; genus ≥ 100 rows are listed for completeness and are infeasible by constants (MMT's cutoff $gl \geq 4097$ kills none of the rows above, but their Table 1 shows how fast T grows with g). The auditor's remedy in every sparse row is the direct magic-number test of the published curve for every factorisation $N = n\cdot l$ (`audit_curve`, `ghs_screen`), which costs polynomial time and which the sparsity of the weak set does not help to evade.

5. Odd characteristic: theorems used (Theorems 1–3)

Theorem 1 (weak-set size of a branch type) [H, checked E]
Fix $n \geq 2$ and a branch type τ: a σ-stable set $S \subset P^1(\bar{F}_q)$ that is a union of σ-orbits of sizes $d_1, \ldots, d_k$, $\Sigma\, d_j = r$, together with the positions of a 4-subset B whose translates $\sigma^i B$ cover S. The branch sets of type τ are obtained by choosing one point of $P^1(F_{q^{d_j}})$ of exact degree d_j per orbit, so there are $\approx \prod q^{d_j} = q^r$ of them (exactly, up to the $\prod d_j$ rotations and the point ∞). The group $PGL_2(F_q)$ (order $q^3 - q$) acts on $P^1(\bar{F}_q)$ commuting with σ, so it maps branch sets of type τ to branch sets of type τ and the cross-ratio class (the j-invariant of the double cover branched at B) is constant on its orbits. A generic stabiliser is trivial, so the number of $j \in F_{q^n}$ carrying a degree-2 function of type τ is $\approx c_\tau\, q^{r-3}$ with c_τ bounded, and a random curve over F_{q^n} is weak of type τ with probability $\approx c_\tau\, q^{r-3-n}$, i.e. $\delta = n + 3 - r$. The literature's densities are the cases $n = 2,\ r = 5$ (all curves: Scholten), $n = 3,\ r = 5$ ($\theta(q^2)$: Momose–Chao), $n = 5,\ r = 5$ ($\theta(q^2)$: Chao 2007 sl. 21), $n = 6,\ r = 6$ ($\leq 1/q^3$: Joux–Vitse App. C) and $n = 7,\ r = 7$ ($\theta(q^4)$: Chao's $\theta(q^{d\ell-3})$). Part B counts the weak j exactly and finds the exponent $r - 3$ on every type (§5).

Theorem 2 (the GHS compositum signature) [T, proved here]
Let $n = 3$, q odd, and let C/F_q be the GHS compositum of a branch type with $(m, r) = (3, 5)$: $S = \{s_0, s_1, s_2\} \cup \{t_1, t_2\}$ with $\{s_i\}$ a 3-orbit, $\{t_1, t_2\}$ σ-stable, $B_i = S \setminus \{s_i\}$. Write $K = F_{q^3}$, $L = K(x, y_0, y_1, y_2)$ with $y_i^2 = f_i(x)$, f_i the quartic with root set B_i, and τ_i the sign change of y_i. Then:

(a) $\iota = \tau_0\tau_1\tau_2$ is the hyperelliptic involution of C_K, and $C/\langle\iota\rangle$ is a conic D. *Proof.* The fixed field of $\langle\iota\rangle$ is $K(x, y_0y_1, y_0y_2)$, a $(Z/2)^2$-cover of P^1_x branched over $B_0 \triangle B_1 = \{s_0, s_1\}$ and $B_0 \triangle B_2 = \{s_0, s_2\}$, three points in all: genus $1 + 2^0 (3 - 4) = 0$ by the formula of Theorem 1 of the previous study. A genus-3 curve with a genus-0 quotient by an involution is hyperelliptic with that involution. ∎

(b) The reduced automorphism group of C_K contains the Klein four-group $V = (Z/2)^3/\langle\iota\rangle$, acting on the hyperelliptic line $D \cong P^1_u$, with quotient P^1_x and branch points s_0, s_1, s_2 (inertia $\tau_j\tau_k$ at s_i); the eight Weierstrass points of C are the two fibres of $D \to P^1_x$ over t_1, t_2, where the inertia is $\langle\iota\rangle$. In a V-adapted coordinate u on D ($V = \{u, -u, 1/u, -1/u\}$, quotient $w = u^2 + u^{-2}$) the Weierstrass octic is $(u^4 - w_1 u^2 + 1)(u^4 - w_2 u^2 + 1)$, so the Weierstrass set is a union of two free V-orbits. The τ_i are K-rational and are permuted cyclically by the q-Frobenius σ (σ: $y_i \mapsto y_{i+1}$), so V is σ-stable and σ acts on its three involutions as a 3-cycle.

(c) Consequently the Frobenius cycle type on the Weierstrass points is $(1,1,3,3)$ when σ fixes each V-orbit (an affine map of a V-torsor of order 3 in S_4 is a 3-cycle with one fixed point) and $(2,6)$ when σ swaps the two orbits. The previous study recorded exactly these two cycle types on all 81 Joux–Vitse instances. ∎

The census tests (b) directly: it finds every F_{q^3}-rational involution of P^1 preserving the Weierstrass set (from the images of three roots), looks for a σ-stable Klein four-group among them whose involutions σ permutes cyclically and which has no fixed point in the Weierstrass set. On all 12 Joux–Vitse controls ($p = 5, 7, 11, 13$, three arms) the signature is found and the cycle type is as stated (test `controls_jv_family`).

Theorem 3 (Weil-restriction-shaped hyperelliptic genus-3 Jacobians)
[T, proved here] Let C/F_q be a hyperelliptic curve of genus 3, q odd, and E/F_{q^3} of trace t. If $Jac(C)$ is isogenous over F_q to $Res_{F_{q^3}/F_q} E$ (by an isogeny of any degree), then t is even, and the q-Frobenius permutes the eight Weierstrass points of C with cycle type $(1,1,3,3)$, $(2,3,3)$, $(1,1,6)$ or $(2,6)$. In particular $\#E(F_{q^3}) = q^3 + 1 - t$ is even: **no hyperelliptic genus-3 Jacobian over F_q is isogenous to the Weil restriction of an elliptic curve of prime order over F_{q^3}.**

Proof. Isogenous abelian varieties have the same characteristic polynomial of Frobenius, $P(T) = T^6 - tT^3 + q^3$, and its reduction modulo 2 is the characteristic polynomial of Frobenius on the 2-torsion $A[2] = T_2(A)/2T_2(A)$. For a hyperelliptic curve with Weierstrass set W ($|W| = 8$), $Jac(C)[2]$ is the Galois module of even-cardinality subsets of W modulo W itself, a subquotient of the permutation module $F_2[W]$; the characteristic polynomial of Frobenius on $F_2[W]$ is $\prod_i (T^{c_i} + 1)$ over the cycle lengths c_i, and the two trivial factors removed by passing to even subsets and modulo W give $\chi_{Jac[2]}(T) = \prod_i (T^{c_i} + 1) / (T + 1)^2$. If t is odd, $P(T) \equiv T^6 + T^3 + 1 = \Phi_9(T) \pmod 2$, irreducible of degree 6 (2 has order 6 modulo 9), and $\Phi_9 \mid T^c + 1$ requires $9 \mid c$, impossible with $c \leq 8$. So t is even and $P(T) \equiv T^6 + 1 = (T + 1)^2 (T^2 + T + 1)^2$. Over F_2, $T^c + 1 = \prod_{d|c} \Phi_d$ with $\Phi_1 = \Phi_2 = T + 1$, $\Phi_3 = T^2 + T + 1$, $\Phi_4 = (T+1)^2$, $\Phi_6 = \Phi_3$, $\Phi_8 = (T+1)^4$, and Φ_5, Φ_7 of degrees 4 and 6 (products of irreducibles different from Φ_3). Hence $\prod_i (T^{c_i} + 1) = (T+1)^4 (T^2+T+1)^2$ forbids cycles of length 5 and 7, needs Φ_3 exactly twice (two 3-cycles, or one 6-cycle), and needs $(T+1)$ with multiplicity 4: with two 3-cycles the remaining two points form $(1,1)$ or (2); with a 6-cycle likewise. That leaves exactly the four cycle types stated. ∎

The census finds **only** these four cycle types among all Res-shaped classes at $q = 3, 5, 7, 11, 13$ (`census_patterns.csv`), and no class with #E prime; the GHS compositums of Theorem 2 occupy the $(1,1,3,3)$ and $(2,6)$ columns, never $(2,3,3)$ or $(1,1,6)$. Theorem 3 gives an auditor a two-line necessary condition (even trace; cycle type) for any claim that a hyperelliptic genus-3 Jacobian over F_q is a Weil-restriction target, and it says that Tian's route ((ℓ,ℓ,ℓ)-isogenies for prime order) must end on non-hyperelliptic curves, for every isogeny and not only for covering maps.

5b. Odd characteristic: exact weak-curve and class counts (Part B)

[E] Setting and notation as in §1 and Theorem 1 ($q = p \in \{3, 5, 7, 11, 13\}$, $n = 2..7$, orbit sizes and bounds as frozen in the protocol; `oddchar_types.csv`, `oddchar_classes.csv`, `oddchar_checks.txt` of run 20261018T0630Z). Every weak j lies in F_{q^n}, every trace is in the Hasse range, the all-j tables have exactly q^n entries, and the $n = 3$ $(3,5)$ curves have $\#E \equiv 0 \pmod 4$ on both twists (checks 5–6, no violation at any (q, n)).

Types found (check 7). Per n the types with genus $\leq g_{max}$ and period n, as ($orbit\ sizes$: m, r, g): $n = 2$: 2+1+1+1, 2+2+1, 3+2 (2, 5, 2); 2+2+1+1, 2+2+2 (2, 6, 3). $n = 3$: 3+1+1, 3+2 (3, 5, 3); 3+3, 6 (2, 6, 3); 3+1+1+1, 3+2+1, 3+3 (3, 6, 5). $n = 4$: 4+1 (4, 5, 5); 4+1+1, 4+2 (3, 6, 5). $n = 5$: 5 (4, 5, 5) only. $n = 6$: 3+2+1, 6 (4, 6, 9). $n = 7$: 7 (3, 7, 7). These are exactly the (m, r, g) triples the previous study's `types.csv` lists at the minimum for each n (`m2r5`; `m2r6, m3r5`; `m3r6, m4r5`; `m4r5`; `m4r6`; `m3r7`), and the $n = 3$ list is Momose–Chao's: 3+1+1/3+2 are the hyperelliptic family (3), 3+3/6 their Types 1 and 2 (degree-4 non-hyperelliptic).

Weak-j counts against q^{r−3} (figures/oddchar_counts.svg). The ratio weak_j / q^{r−3} at q = 7, 11, 13:

n	type (m, r, g)	q = 7	q = 11	q = 13	δ = n + 3 − r	a = δ − n/2
2	2+1+1+1 (2,5,2)	0.143	0.157	0.154	0	−1
2	3+2 (2,5,2)	0.286	0.306	0.308	0	−1
3	3+1+1 (3,5,3)	0.510	0.496	0.485	1	−0.5
3	3+2 (3,5,3)	0.429	0.455	0.462	1	−0.5
3	3+3 (2,6,3)	0.169	0.166	0.165	0	−1.5
3	6 (2,6,3)	0.169	0.166	0.167	0	−1.5
3	3+2+1 (3,6,5)	0.490	0.496	0.497	0	−1.5
4	4+1 (4,5,5)	1.000	1.041	1.018	2	0
4	4+1+1 (3,6,5)	0.499	0.515	0.505	1	−1
4	4+2 (3,6,5)	0.429	0.455	0.462	1	−1
5	**5 (4,5,5)**	**1.020**	**1.000**	**1.006**	**3**	**+0.5**
6	3+2+1 (4,6,9)	0.980	0.992	0.994	3	0
6	6 (4,6,9)	0.974	0.994	0.993	3	0
7	7 (3,7,7)	1.992 (q=7), 2.005 (q=5), 1.815 (q=3)			3	−0.5

Every ratio is within a factor of 1.2 of itself across the three largest q (the decision rule asked for a factor 3): **H_B holds for every type enumerated**, with constants c_τ between 0.14 and 2.0. The exact counts at q = 13: n = 5 has 170 weak j among 371293 (1/q^3); n = 3 has 82 + 78 hyperelliptic-family and 363 + 366 Type 1/2 weak j among 2197 (about 0.4 q^2, Chao's ≈ q^3/2 Legendre forms counted per j and per type); n = 6 has 2184 + 2181 among 4826809 (≈ q^3, Joux–Vitse's "at most 1/q^3" met with constant 1); n = 7 at q = 7 has 4782 among 823543 (2 q^4).

Isogeny-class statistics (oddchar_classes.csv, all-j trace tables for q^{2n} ≤ 2·10^11; figures/oddchar_classes.svg). Share of all curves over F_{q^n} that lie in a class containing a weak curve of the type, and the mean number of weak curves per covered class:

n	type	share q = 7	q = 11	q = 13	mean weak per covered class (7, 11, 13)
2	2+1+1+1	0.41	0.42	0.42	2.3, 3.2, 3.7
3	3+1+1	0.25	0.25	0.22	5.6, 7.5, 8.2
3	3+2+1 (g=5)	0.66	0.66	0.67	10.5, 20.0, 25.4
3	3+3 (g=3, Type 1)	0.42	0.42	0.42	6.8, 12.6, 17.3
4	4+1	0.15	0.18	0.13	7.0, 6.3, 8.6
4	4+1+1	0.39	0.38	0.23	9.5, 14.6, 28.5
5	**5**	**0.077**	**0.061**	**0.054**	**7.1, 6.7, 6.1**
6	6 (q = 7 only)	0.14	—	—	7.8
7	7 (q = 3, 5)	0.34, 0.34	—	—	12.3, 15.7

Reading. (i) At n = 3 the covered share of the hyperelliptic family saturates near 0.25 while the number of weak curves per covered class grows (5.6 → 8.2): the weak curves pile up in the classes with #E ≡ 0 (mod 4) (check 5), exactly the restriction Joux–Vitse state ("not all curves are vulnerable to this isogeny walk") and Yee quantifies ("around half of the ordinary isogeny classes will not contain any weak curves"); the Type-1 covers (3+3) reach 0.42, and the genus-5 type 0.67. (ii) At n = 5 the covered share **falls** with q (0.077 → 0.054, the ratio 0.70 against the predicted (7/13)^{1/2} = 0.73 for q^{n/2−δ} = q^{−1/2} weak members per class) while the weak count per covered class stays flat: this is the sparse regime, where most classes hold no weak member and a planted one is not reached by a walk. (iii) At n = 4 the r = 5 type (a = 0) keeps a flat share and the r = 6 types a larger, drifting one, as the margins say. (iv) The walk length inside a covered class (curves over weak curves) grows with q for every n ≥ 4 (n = 5: 26, 81, 118), roughly as 0.06 q^3 against rho's q^{5/2} (130, 401, 609): at toy size the ratio is flat (0.20, 0.20, 0.19) because the constant is small; the exponent gap q^{1/2} only shows beyond q ≈ 250. Nothing at q ≤ 13 is a statement about cryptographic sizes; the direction of every drift matches §1.

Where the odd-characteristic Teske field is. n = 5 is the only extension degree with a positive margin: the family y^2 = ∏_{i=0}^{3}(x − α^{q^i}), α of degree 5 (Hess Table 3, Diem 2003), has ≈ q^2 curves among q^5, descends to genus 5 over F_q (owner exponent 2 − 2/5 = 1.6 against rho's 2.5), and a walk reaches it in ≈ q^3 > q^{5/2} steps. The test an auditor needs is the one Yee's thesis asks for in its last paragraph: given E/F_{q^5}, decide whether some degree-2 function has a branch set of this type. The branch set is a 4-subset of a 5-orbit, so the test is: over the PGL_2(F_q)-orbit representatives of the q^5-many degree-5 α (there are ≈ q^2), check whether j(E) equals the cross-ratio invariant of {α, α^q, α^{q^2}, α^{q^3}}; Part B's enumeration is this test run for every E at once. At n = 6 the margin is 0 with genus 9 (Joux–Vitse's App. C family), and at n = 4 the r = 5 type sits at margin 0 with genus 5.

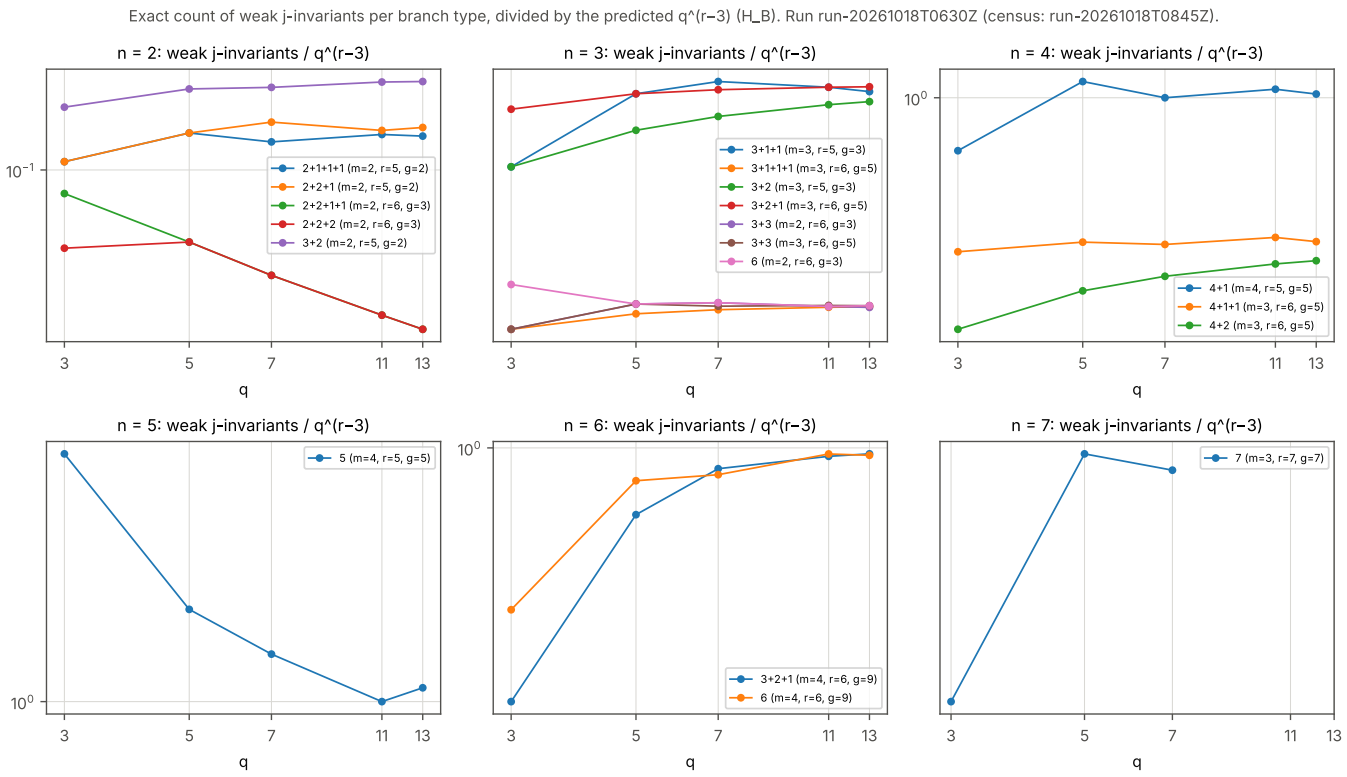


Figure 3: Weak j-invariants per branch type divided by q^(r−3) (figures/oddchar_counts.svg, from oddchar_types.csv).

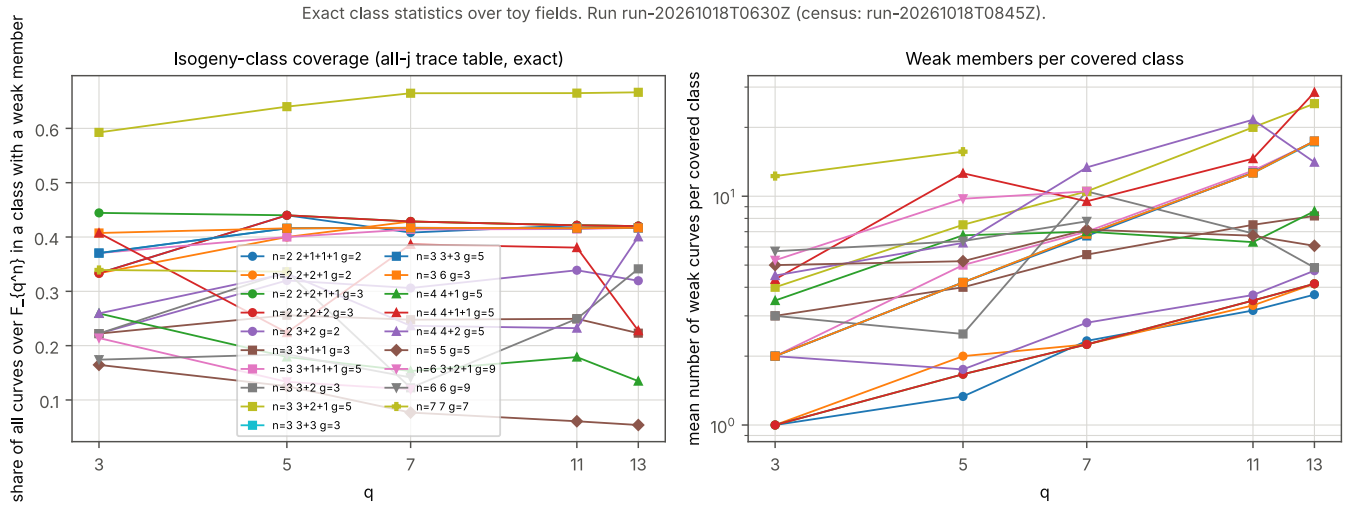


Figure 4: Isogeny-class coverage and weak members per covered class (figures/oddchar_classes.svg, from oddchar_classes.csv).

6. The census: Weil-restriction-shaped genus-3 hyperelliptic

Jacobians (Part C) [E, T] Run `20261018T0845Z` (`census.csv`, `census_summary.csv`, `census_patterns.csv`, `census_checks.txt`; the `q ≤ 11` files are byte-identical to those of run `20261018T0630Z`, which was killed at `q = 13`). For every odd `q ≤ 13` the tool enumerates every squarefree `y^2 = c f(x)` (`c ∈ {1, ν}`, `f` monic of degree 8 with no `x^7` term or of degree 7 with no `x^6` term), keeps the Res-shaped ones (`#C(F_q) = q + 1`, `#C(F_{q^2}) = q^2 + 1`), computes `t` from `#C(F_{q^3})`, joins models into `F_q`-isomorphism classes by the moves that generate `PGL_2(F_q)` modulo translations (every neighbour found, no duplicate model: asserted), and records per class the trace, the Weierstrass cycle type, the `F_{q^3}`-rational reduced automorphism group and the GHS signature of Theorem 2. Checks 8–9 pass at every `q` (no Hasse violation, `t` constant on each class, every signature class has `#E ≡ 0 (mod 4)` and cycle type `(1,1,3,3)` or `(2,6)`); the twelve Joux–Vitse controls pass (check 10).

q	models	square-free	Res-shaped models	classes	signa-ture	no sig-nature	(1,1,3,3) yes / no	(2,6) yes / no	(1,1,6)	(2,3,3)	#E prime	t sub-field-shaped (with sig.)	traces (with sig.)
3	5,832	3,888	152	21	6	15	4 / 2	2 / 2	11	0	0	7 (0)	9 (2)
5	187,500	150,000	3,088	158	20	138	12 / 30	8 / 20	76	12	0	70 (8)	21 (10)
7	3,294,172	2,823,576	48,352	538	39	499	19 / 116	20 / 134	211	38	0	92 (1)	37 (9)
11	42,517,464	38,652,240	293,440	3173	106	3067	54 / 832	52 / 803	1168	264	0	547 (6)	73 (16)

13	135,150,652	124,754,448	682,472	5582	160	5422	92 / 1264	68 / 1432	2198	528	0	1042 (28)	93 (36)

Reading. (i) **Theorem 3 is met exactly**: among all 9,472 Res-shaped classes only the four predicted cycle types occur, no other pattern (`8`, `1+7`, `3+5`, `4+4`, `2+2+4`, …) appears although squarefree octics of every pattern were enumerated, and no class has `#E` prime (`classes_numE_prime = 0` at every `q`). (ii) **H_C is refuted**: 331 of 9,472 classes carry the compositum signature; at `q = 13`, `160` of `5582`. The signature classes grow like `q^2` (`6, 20, 39, 106, 160` at `q = 3, 5, 7, 11, 13`), in step with the number of weak `j` of the `n = 3` hyperelliptic family in Part B (`6, 22, 46, 115, 160`: at `q = 13` the two counts coincide), while the non-signature classes grow like `q^{3.5}` (`15, 138, 499, 3067, 5422`; `5422/3067 = 1.77` against `(13/11)^{3.5} = 1.79`), the growth expected if a positive fraction of all ≈ `q^5` genus-3 hyperelliptic curves have `a_1 = a_2 = 0` (`a_1 ∈ O(√q)`, `a_2 ∈ O(q)`). (iii) The non-signature classes split by cycle type: the `(1,1,6)` and `(2,3,3)` ones (`2198 + 528` at `q = 13`) can never be compositums by Theorem 2(c); the `(1,1,3,3)` and `(2,6)` ones (`1264 + 1432`) pass the necessary cycle-type condition but have no Frobenius-cyclic Klein four-group: their `F_{q^3}`-rational reduced automorphism group is trivial for `1202` of the `1264` `(1,1,3,3)` classes and of order 2 for the other `62`, and no class has a σ-stable Klein four-group without also having the signature. (iv) `t` has the subfield shape `t_0^3 − 3qt_0` (so `Res E ~ E_0 × T` splits) for `1042` classes at `q = 13`, `28` of them with the signature; the remaining `4380` non-signature classes are isogenous to a *simple* Weil restriction by an isogeny that is not the compositum's. (v) Every signature class has `#E ≡ 0 (mod 4)`, as Joux–Vitse state for the family (3), and the traces reached by signature classes (`36` of `93` at `q = 13`) are a minority of the traces reached by Res-shaped Jacobians at all.

What this says for the search. The association `Jac(C) ~ Res E` is a property of the isogeny class of `E` (its trace), not of a model, so it is not hideable; what varies with the model is the *degree* of the cover `C_K → E` induced by the isogeny, which the census does not compute. The compositum classes have a known degree-8 cover; for the `5422` others at `q = 13` the degree is open [O]. The strong form of the question, whether a non-compositum class carries a cover of degree comparable to 8, is the one experiment that would widen the odd-characteristic weak set beyond Momose–Chao's families, and §7 names it as the next step.

Exact census (every squarefree octic and septic over F_q). Run run-20261018T0630Z (census: run-20261018T0845Z).

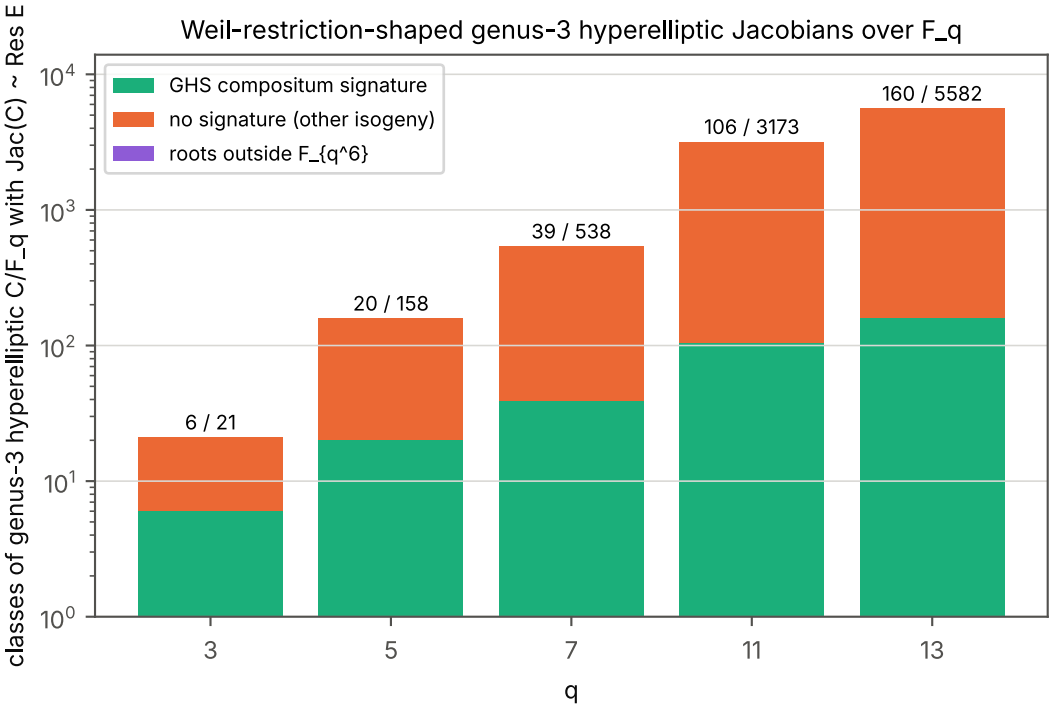


Figure 5: Res-shaped genus-3 hyperelliptic classes by signature (figures/census.svg, from census_summary.csv).

7. Other ways to attach a Weil-descent target to a curve, and where

to look next Ways found in the literature and in this round, with their class-invariance:

association	field	what it needs	class invariant?	hideable?	source
GHS Artin–Schreier compositum	`F_{2^{nl}}`	small `deg Ord_b`	no (`b`)	yes when `a > 0` (§4)	GHS 2002; MMT 2002
gGHS (Hess)	`F_{2^{nl}}`	`b = (αβ)^2` with small annihilators	no	mostly dense (§4)	Hess 2003/2004
Kummer compositum	`F_{q^n}`, `q` odd	branch type with small `(m, r)`	no (branch set)	only `n = 5` (§5)	Diem 2003; Momose–Chao
isogeny-class extension	both	a weak member in the class	by definition	the walk is the auditor's tool, cost `q^{min(6,n/2)}`	GHS 2002 §4; Yee 2016
`Jac(C) ~ Res E` by L-polynomial (Tate)	`F_{q^n}`	`L_C = L_{Res E}`	yes (isogeny class of `Res E`)	no: it is a property of the trace `t`	Part C; Theorem 3
`(ℓ,ℓ,ℓ)`-isogeny `Res E → Jac(C)`	`F_{q^3}`	a rational maximal isotropic `Λ ⊂ Res E[ℓ]`	class-level existence; curve-level construction	not studied [O]	Tian 2020

Scholten form, genus 2 over F_{q^2}	F_{q^4}, F_{q^2}	full rational 2-torsion, or odd order and j ∉ F_q	partly (2-torsion is a volcano-level property)	no: 3/4 of curves	Scholten; Joux–Vitse §5
trace-zero / decomposition index calculus	F_{q^n}	nothing special	yes (group only)	no: everyone	Gorla–Massierer; Gaudry–Diem
C_{ab} curves on the Weil restriction, char 3	F_{3^n}	composite n, special E	no	not assessed [O]	Arita 2000

Where a "different type of curve" could still hide, and the discriminating step for each:

1. **F_{q^5} in odd characteristic.** The genus-5 family y^2 = ∏_{i=0}^{3}(x − α^{q^i}) is sparse (a = +1/2) and its branch type is the only one with genus ≤ 9 and period 5 (Part B found exactly one type for n = 5 at every q). The auditor's test is explicit: for a published E/F_{q^5}, decide whether some degree-2 function has branch set {α, …, α^{q^3}} for a degree-5 α; Yee's open question (an efficient weak-form test in odd characteristic) is exactly this, and the Legendre-form criterion of Chao sl. 29–32 (a quadratic condition in λ) is the model to follow. Next step: write the n = 5 analogue of ghs_screen (the B-test over PGL_2(F_q)-orbits).
2. **Non-GHS Res-shaped Jacobians.** The census shows that, already at genus 3, most hyperelliptic Jacobians isogenous to a Weil restriction are not compositums. Each is a target with a cover of *some* degree; the degree decides whether the transfer is computable. Next step: for the non-signature classes at q = 5, 7, compute the kernel of an explicit isogeny Res E → Jac(C) (via Tian's normal-Weil-set method for ℓ = 3, or by matching (2,2,2)-kernels) and record its degree.
3. **The (1,1,6) and (2,3,3) columns.** By Theorem 3 these Jacobians are isogenous to Res E but can never be GHS compositums. A constant-fraction count of them against the (1,1,3,3)/(2,6) columns at every q (§6) says the non-compositum relation is not an artefact of small q.
4. **Binary n = 21, 35.** Hess's pairing gives genus 21 at n = 21 and 105 at n = 35 from products of non-primitive factors, sparser than the basic descent (δ = 17 against 16). Whether the gGHS map is explicit there, and the Enge–Gaudry cost at genus 21 over F_{2^8} (N = 168) or F_{2^10} (N = 210), is a direct extension of MMT's table with Hess's genus formula. These N are not in use; the row is an auditor's checklist entry, not a threat.
5. **Characteristic 3.** Odd-characteristic Kummer descents apply verbatim (Theorem 1 does not see q), so F_{3^{5l}} has the same a = +1/2 regime; Arita's C_{ab} route is a second, Artin–Schreier flavoured, association that this round did not enumerate [O].

Negative results of the round, kept: no branch type of period n with genus below the previous study's floor was found (check 7); no Res-shaped hyperelliptic Jacobian with prime #E exists (Theorem 3, confirmed by the census at five q); no prime-field variant exists via descent (§3).

8. Checks and controls

check (PROTOCOL.md)	where	outcome
1. n = 7: type I (3, 7, δ=4), type II (4, 8, δ=3), gGHS (3+3, 7, δ=2); GHS 2002 per-class table 0.404, 0.286, 0.269, 0.286, 0.323	test check1_n7	pass
2. n = 31, q = 2^5: basic δ = 25, per class ≈ 2^{−52}; gGHS (5,5): genus 31, δ = 22, ≈ 2^{−38}	test check2_n31	pass
3. MMT counts 2^94, 2^57, 2^171 for N = 161, 165, 170	test check3_mmt_counts	pass
4. n = 15: minimal m = 5; n = 21: m = 6 from a degree-5 product	test check4_composite_n	pass
5. every j(B) ∈ F_{q^n}; every trace in the Hasse range; n = 3, (3,5): #E ≡ 0 (mod 4) on both twists	oddchar_checks.txt	pass (0 violations at every (q, n))
6. all-j tables list exactly q^n values	oddchar_checks.txt	pass (25 tables, counts equal q^n)
7. branch types found per n against types.csv of the previous study	this report, §5	pass (§5b)
8. #C(F_{q^3}) = q^3 + 1 − 3t with t integral in the Hasse range; t constant on each class	census_checks.txt	pass (0 violations, t constant on every class, 5 q)
9. signature classes have #E ≡ 0 (mod 4) and cycle type (1,1,3,3) or (2,6)	census_checks.txt	pass (331 signature classes)
10. the 12 Joux–Vitse controls (p = 5, 7, 11, 13; linear, split, conjugate) are Res-shaped with the recorded t, cycle type and signature	test controls_jv_family	pass
field tables: table addition equals digit addition; inverses; subfield membership	test field_tables_consistent	pass
census bookkeeping: no duplicate model, every neighbour of a Res-shaped model is in the enumeration (asserted)	census_checks.txt, assertions	pass (0 missing at every q)

Negative and positive controls. The twelve Joux–Vitse H curves are the positive control for the signature test (all pass; a false test would have had to pass them by accident with the Klein-four group found from the roots). The negative control is Theorem 3: a class with a cycle type outside the four allowed ones, or with #E prime, would contradict a proof; none appeared among 9,472 classes. The exponent test of H_B is controlled by the literature's densities at n = 2, 3, 5, 6, 7 (Theorem 1), which the counts reproduce.

9. What is not established

- [O] No cost here is measured. The exponents `n/2`, `2 − 2/g` and `min(δ, n/2)` drop constants that decide feasibility in practice (MMT 2002 Table 1 has the Enge–Gaudry constants for binary fields; for genus `≥ 100` nothing is feasible).
- [H] The uniform distribution of weak curves over isogeny classes is the heuristic of GHS 2002 §5; it is tested only at toy size (Yee 2016 and Part B here), and Sato–Tate makes classes unequal in size. Part B's class statistics are exact at `q ≤ 13` and show the predicted direction of drift, not an asymptotic law.
- [O] The census does not compute the degree of the isogeny `Res E → Jac(C)` for the non-signature classes; "isogenous" says a cover of some degree exists, not that it is computable.
- [O] Part B enumerates branch types with orbits inside one ambient field per `n` (sizes `{1,5}` at `n = 5`, `{1,7}` at `n = 7`, no 3-orbits at `n = 4`); types with other orbit sizes have `r ≥ 7` at those `n` and genus above the frozen `g_max`, so the smallest-genus types are not affected, but the count of larger-genus types is incomplete there.
- [O] Characteristic 3 and Arita's `C_{ab}` route, Thériault's superelliptic (`y^d = f`) descents, and the non-hyperelliptic genus-3 census (plane quartics) are not enumerated.
- [O] Nothing here concerns any standardised curve: the NIST and SEC binary curves use prime `N`, the prime-field curves have no descent (§3), and `F_{2^155}`, `F_{2^161}` and `F_{q^5}` fields are discussed only as the literature's examples.
- [O] No Lean statement accompanies Theorems 2 and 3 (AGENTS.md "Mathematical results should include Lean proofs"); they are elementary (Riemann–Hurwitz, the `F_2` factorisation of `T^c + 1`), and a formalisation is listed as owed.

10. Canonical graphs checked

No registered curve gains or loses a route, so `docs/curves/jacobian-transfers.json`, `docs/curves/JACOBIAN_TRANSFERS.md`, `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json` and the `docs/performance-gains/` graphs are unchanged. The `GHS_GENUS_FLOOR` table in `src/cryptanalysis/jacobian_transfer/mod.rs` (`[1, 2, 3, 5, 5, 9, 7, …]`) is consistent with the smallest genera found here (`n = 2..7`: `2, 3, 5, 5, 9, 7`). The previous study's `figures/genus_vs_n.svg` is unchanged. New graphs of this round: `figures/margins.svg`, `figures/oddchar_counts.svg`, `figures/oddchar_classes.svg`, `figures/census.svg`, and the diagram `figures/taxonomy.svg`.

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